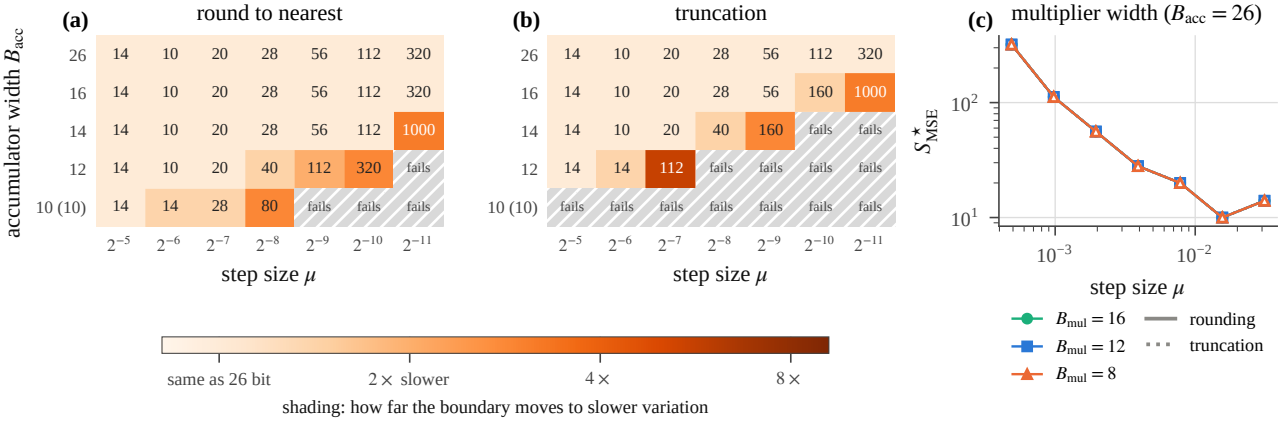
(a)
round to nearest
(b)
truncation
(c)
multiplier width ($B_{\text{acc}} = 26$)
accumulator width B_{acc}
26 14 10 20 28 56 112 320
16 14 10 20 28 56 112 320
14 14 10 20 28 56 112 1000
12 14 10 20 40 112 320 fails
10 (10) 14 14 28 80 fails fails fails
26 14 10 20 28 56 112 320
16 14 10 20 28 56 160 1000
14 14 10 20 40 160 fails fails
12 14 14 112 fails fails fails fails
10 (10) fails fails fails fails fails fails fails
2^{-5} 2^{-6} 2^{-7} 2^{-8} 2^{-9} 2^{-10} 2^{-11}
step size μ
$S^{\star}_{\text{MSE}}$
10^2
10^1
10^{-3}
10^{-2}
step size μ
$B_{\text{mul}} = 16$
$B_{\text{mul}} = 12$
$B_{\text{mul}} = 8$
rounding
truncation
same as 26 bit
2× slower
4×
8×
shading: how far the boundary moves to slower variation